NOTE To simplify the coding exercises, I've created a ch04_mapping.txt file under the kibana_scripts folder at the root of the book's repository (http://mng.bz/OxXo). Copy the contents of this file to Kibana as is. You can work through the examples by executing the individual code snippets while following the chapter's content.

4.1.1 Mapping definition

Every document consists of a set of fields representing business data, and every field has one or more specific data types associated with it. The *mapping definition* is the schema of the fields and their data types in a document. Each field is stored and indexed in a specific way based on data type. This helps Elasticsearch support many search queries such as full-text, fuzzy, term, and geo.

In programming languages, we represent data with specific data types (strings, dates, numbers, objects, etc.). It is a common practice to let the system know the types of variables during compilation. In a relational database, we define a table schema with appropriate field definitions to persist the records, and it is mandatory for the schema to exist before we start our persistence in the database.

Elasticsearch understands the data types of the fields while indexing documents and stores the fields into appropriate data structures (for example, an inverted index for text fields and block k-dimensional (BKD) trees for numerical fields) for data retrieval. Indexed data with precisely formed data types leads to accurate search results and helps sort and aggregate the data.

Figure 4.1 shows the anatomy of the mapping schema for an index. As you can see, the index consists of a `mapping` object made up of individual fields and their types enclosed in a `properties` object.

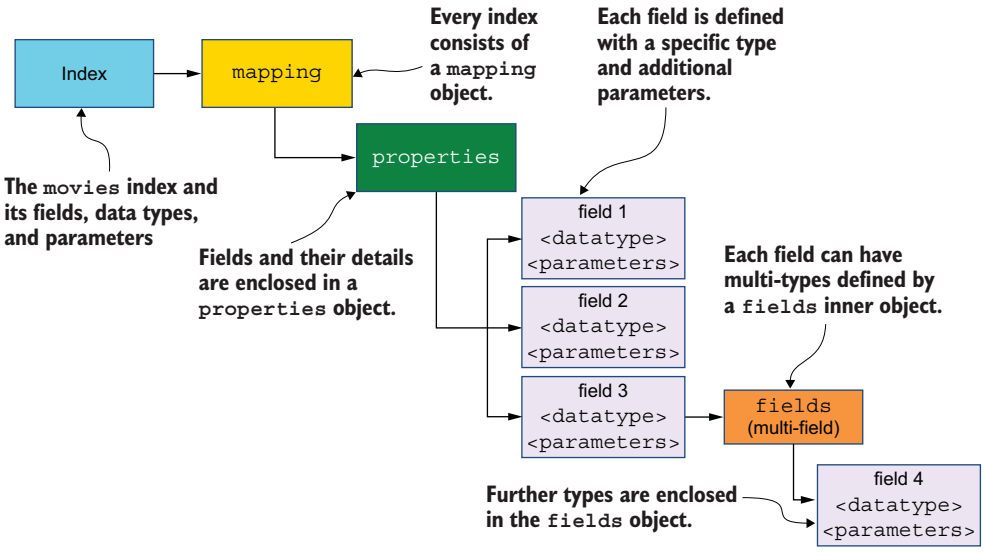


Figure 4.1 Anatomy of a mapping schema